

Neutron Transport and Tritium Breeding Modelling in Nuclear Fusion Reactor Breeder Blankets

MPhys Project Report

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Abstract

The pursuit of practical nuclear fusion is focused on the deuterium-tritium (D-T) fuel cycle, widely considered the most feasible pathway for near-term power generation. A critical challenge, however, is the reliance on tritium, a radioactive isotope with an extremely limited external supply, necessitating a self-sufficient fuel cycle. The D-T fusion reaction produces most of its energy as 14.1 MeV neutrons, which can in turn be used to regenerate tritium through neutron-lithium interactions. The reactor subsystem that performs this function—the breeder blanket—is therefore essential for the viability of the D-T fuel cycle. It must simultaneously extract heat, shield reactor components, and maintain tritium self-sufficiency. This project applies Monte Carlo neutron transport simulations using **OpenMC** to investigate the neutronic performance of advanced breeder blanket designs, including heterogeneous and functionally graded geometries. The study aims to quantify how variations in material composition and spatial structure influence the tritium breeding ratio (TBR) and neutron flux profiles, and explores the potential for automated optimisation of blanket parameters through computational frameworks. The outcomes will inform the feasibility and efficiency of next-generation blanket concepts relevant to future magnetic confinement fusion reactors such as DEMO and STEP.

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1. INTRODUCTION

The pursuit of practical nuclear fusion represents one of the most ambitious and potentially transformative scientific endeavours of the twenty-first century. As global energy demand continues to rise and the environmental impacts of fossil fuel use become ever clearer, fusion offers the prospect of a near-limitless, low-carbon energy source. Fusion energy aims to replicate, on Earth, the processes that power the Sun — combining light nuclei to release energy through mass-energy conversion. Unlike conventional nuclear fission, fusion promises inherent safety, reduced long-lived radioactive waste, and abundant fuel sources.

Among the various fusion reactions studied, the deuterium-tritium (D-T) reaction has emerged as the most viable for near-term applications due to its relatively high cross-section at achievable plasma temperatures and its favourable energy yield. However, the practical realisation of a D-T reactor introduces challenges that extend far beyond plasma confinement and heating. Chief among these is the supply of tritium, a radioactive isotope that does not occur naturally in significant quantities and must be produced artificially.

This challenge has led to the development of the breeder blanket concept — an integrated structure surrounding the plasma that captures energetic neutrons produced in fusion reactions and uses them to generate tritium through interactions with lithium. In addition to tritium production, breeder blankets also serve several critical roles: moderating neutron flux, capturing energy for conversion to electricity, and protecting reactor components from radiation damage.

Understanding the behaviour of neutrons within the blanket region is therefore central to the success of any D-T fusion system. Computational modelling of neutron transport, energy deposition, and tritium breeding provides vital insights into how different blanket materials and geometries perform under fusion-like conditions. These neutronics studies inform the design and optimisation of test blanket modules and full-scale reactor concepts.

This project contributes to that effort by using advanced Monte Carlo simulations to model neutron interactions within candidate breeder blanket materials. Through this, it aims to identify promising compositions and geometries capable of sustaining tritium self-sufficiency in future fusion power plants.

2. THEORY

2.1. Fundamentals of Nuclear Fusion

Nuclear fusion is the process by which two light atomic nuclei combine, or “fuse,” to form a single, heavier nucleus [1]. This reaction is the primary process powering stars, and it is a candidate for future terrestrial energy generation.

The release of energy in fusion is a direct consequence of **nuclear binding energy**. The nucleons (protons and neutrons) in a nucleus are held together by the strong nuclear force. The binding energy is the energy required to disassemble the nucleus into its separate components. Conversely, when a nucleus is formed, this amount of energy is released.

Crucially, the binding energy *per nucleon* is not constant across all elements. Lighter nuclei,

such as hydrogen isotopes, are relatively loosely bound. Nuclei in the “medium-mass” range, like helium-4 or iron, are much more tightly bound.

Therefore, when two light nuclei fuse to form a heavier, more tightly bound nucleus, the total binding energy of the system *increases*. This surplus energy is released during the reaction. This gain in binding energy corresponds to a reduction in the total rest mass of the system. This difference, known as the **mass defect** (Δm), is not truly “lost” but is converted into energy (E) according to Einstein’s principle of mass-energy equivalence [1].

This relationship is famously expressed by the equation $E = mc^2$, or more specifically for a reaction, as a release of energy Q (the “Q-value”):

$$Q = \left(\sum m_{\text{reactants}} - \sum m_{\text{products}} \right) c^2 = (\Delta m) c^2 \quad (1)$$

Because the speed of light squared (c^2) is an enormous constant, even a minuscule decrease in mass releases a tremendous amount of energy, typically in the form of kinetic energy of the products (e.g., neutrons, alpha particles) and/or high-energy photons (gamma rays).

For terrestrial power generation, the most promising reaction is that between the two heavy isotopes of hydrogen, deuterium (d) and tritium (t) [2, 3]:



This D-T reaction is the primary focus of mainstream fusion research, as it has a fusion cross-section significantly higher than other candidate reactions (such as D-D or D- ^3He) (Fig. 1) [3]. This is mainly due to the stability of the final products linking to a higher Q value output [1].

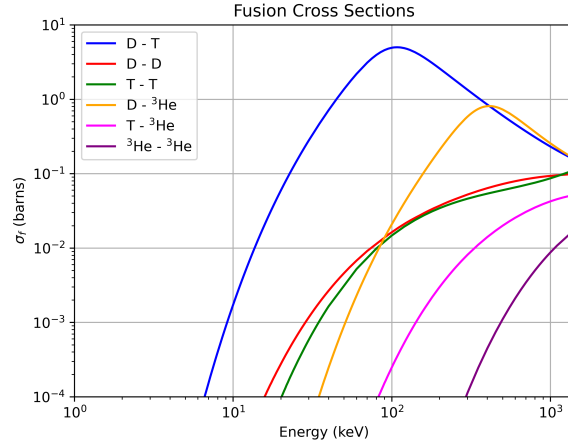


Figure 1: A comparison of fusion reaction reactivities total cross section σ as a function of ion temperature (keV) for the D-T, D-D, and D- ^3He fuel cycles. The D-T reaction’s cross section peaks at a lower temperature and is significantly larger than the alternatives, making it the most accessible reaction for current and near-term fusion devices.

Another important reason for D-T is the comparatively lower required input energy / temperature. To initiate fusion, the positively charged nuclei must be given enough kinetic energy to overcome their mutual electrostatic repulsion, known as the Coulomb barrier. This barrier will increase if the reactants have a more positive charge and so more electrostatic

repulsion. This can exclude cycles like $D-^3He$ as the required temperatures/energies are already high enough (hotter than the core of the sun). In Magnetic Confinement Fusion (MCF) devices, such as the **tokamak** [4], this is achieved by heating the D-T gas into a plasma state at temperatures exceeding 100 million Kelvin. The hot plasma is then confined by powerful magnetic fields, preventing it from touching the reactor walls. The 17.6 MeV of energy released is partitioned between the two products: the alpha particle (α , a helium nucleus) carries 3.5 MeV, while the neutron (n) carries 14.1 MeV [3].

This energy split is fundamental to the reactor’s operation. The charged alpha particle remains trapped by the magnetic fields, depositing its energy into the plasma and helping to sustain its high temperature (a process known as “plasma burning”). The neutron, being electrically neutral, is un-bothered by the magnetic fields and escapes the plasma immediately, carrying 80% of the fusion energy with it [3].

2.2. Neutron Transport and Interactions

The 14.1 MeV neutron is the primary vehicle for both energy extraction and fuel production. Once it leaves the plasma, it travels into the surrounding structures, chiefly the **breeder blanket**. The study of its journey and interactions with the blanket materials is the domain of **neutron transport** [5].

As the neutron moves through matter, it interacts with atomic nuclei via two main processes: scattering and absorption. The probability of any specific interaction occurring is defined by the material’s **microscopic cross-section** (σ), a value that is highly dependent on the energy of the incident neutron [5].

While σ represents the effective target area of a single nucleus, reactor engineering requires understanding interactions within bulk matter. The probability of interaction per unit path length is defined as the **macroscopic cross-section** (Σ):

$$\Sigma = N\sigma \quad (2)$$

where N is the atomic number density of the material [1]. Consequently, the average distance a neutron travels through a medium before undergoing an interaction, known as the **mean free path** (λ), is inversely proportional to the total macroscopic cross-section:

$$\lambda = \frac{1}{\Sigma_{total}} \quad (3)$$

This parameter is fundamental to blanket design, as it dictates the physical thickness required to effectively attenuate the neutron flux and capture energy [1].

These cross-sections are meticulously measured and compiled in comprehensive libraries such as the Evaluated Nuclear Data File (ENDF) [6].

Key interactions within the blanket include:

2.2.1. Scattering Interactions

The neutron collides with a nucleus and “bounces” off, transferring a portion of its kinetic energy to the nucleus. In **elastic scattering** (n, n'), kinetic energy is conserved. In **inelastic**

scattering (n, n'), the neutron excites the nucleus, which then de-excites by emitting a gamma ray, resulting in a larger energy loss for the neutron. Both processes are crucial for **moderation**—slowing the fast 14.1 MeV neutrons down. This thermal energy, deposited in the blanket material, is what is ultimately extracted by a coolant to generate electricity.

2.2.2. Absorption Interactions

The neutron is captured by a nucleus. This can be a **radiative capture** (n, γ) event, where the nucleus emits a gamma ray. This is often a parasitic reaction, as it removes a neutron from the system that could have been used for breeding. Alternatively, the absorption can induce **charged particle emission**, such as (n, p) or (n, α) reactions. This is the fundamental mechanism used for tritium breeding [5].

2.3. Tritium Breeding and Neutron Multiplication

The primary nuclear function of the blanket is to use the fusion neutrons to “breed” new tritium fuel. This is accomplished by bombarding lithium isotopes with the neutrons. There are two reactions that produce tritium:



The ${}^6\text{Li}$ reaction is exothermic and has a very large cross-section for low-energy (thermal) neutrons, making it the primary breeding reaction of interest [7]. The ${}^7\text{Li}$ reaction is endothermic, requiring high-energy neutrons (a “threshold” reaction), and is generally less effective.

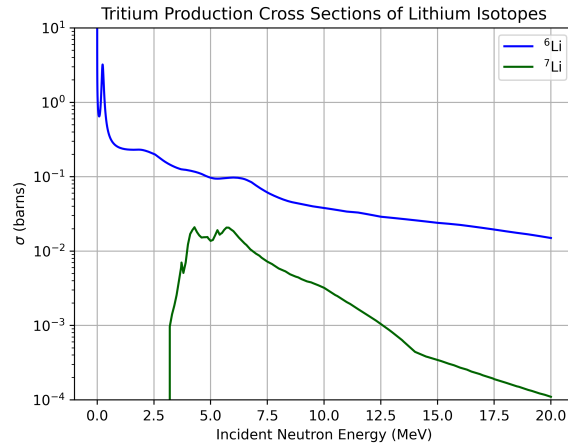


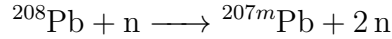
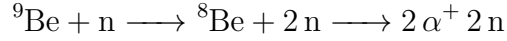
Figure 2: Neutron cross-sections for the primary tritium breeding reactions, ${}^6\text{Li}(n, \alpha)t$ and ${}^7\text{Li}(n, \alpha)t + n'$, as a function of incident neutron energy. The ${}^6\text{Li}$ cross-section is dominant at low (thermal) energies, while the ${}^7\text{Li}$ reaction is a threshold reaction requiring high-energy neutrons.

The critical metric for the blanket’s success is the **Tritium Breeding Ratio (TBR)**, defined as the ratio of tritium atoms produced in the blanket to the tritium atoms consumed in the plasma [8]. Due to inevitable fuel cycle losses (e.g., incomplete plasma burn-up, decay during

extraction), the blanket must breed more tritium than is consumed. To achieve a self-sufficient fuel cycle, a TBR > 1.0 is mandatory, with most reactor designs targeting a value of 1.1 or higher [9, 10].

This presents a “neutron economy” challenge. The D-T reaction produces one neutron, and the primary ${}^6\text{Li}$ breeding reaction consumes one neutron [11]. This leaves no margin for neutrons that are inevitably lost, either by leaking out of the blanket or through parasitic absorption in structural materials (like steel) [11].

To overcome this deficit, the blanket must include a **neutron multiplier** [12, 7]. These are materials that undergo an **(n, 2n)** reaction, where one high-energy incident neutron strikes a nucleus and causes two neutrons to be emitted. The two most viable multiplier materials are Beryllium (Be) and Lead (Pb) [7].



By incorporating these materials, the single 14.1 MeV fusion neutron can be multiplied into two or more lower-energy neutrons. These neutrons are then moderated (slowed down) within the blanket until they are at the optimal thermal energy to be captured by ${}^6\text{Li}$, thus enabling a TBR significantly greater than one [7].

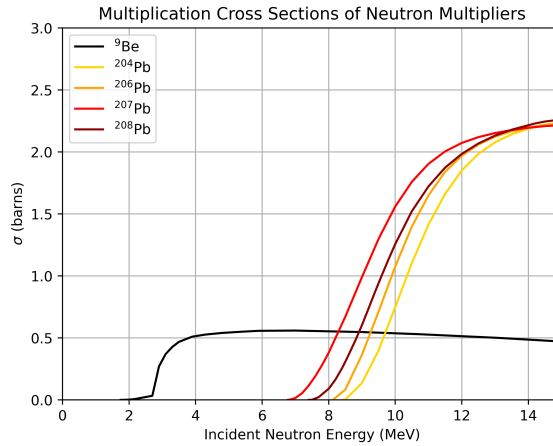


Figure 3: Neutron cross-sections for the (n, 2n) neutron multiplication reactions in Beryllium (${}^9\text{Be}$) and Lead (${}^{208}\text{Pb}$). Both are threshold reactions that require high-energy neutrons, such as those from the D-T fusion reaction (14.1 MeV), to effectively multiply the neutron population.

3. LITERATURE REVIEW

3.1. The D-T Reaction and the Tritium Fuel Imperative

The deuterium-tritium (D-T) fusion reaction, $\text{d} + \text{t} \longrightarrow \text{n} + \alpha + 17.6 \text{ MeV}$, has been the leading candidate for first-generation fusion power plants since its potential was identified in the 1940s [2]. Its primary advantage lies in its high fusion cross-section (σ_f) at plasma temperatures

achievable with current technology, significantly higher than other candidates like D-D or D- ^3He reactions. Furthermore, 80% of the energy (14.1 MeV) is carried by the emitted neutron, which allows for energy to be efficiently extracted from the plasma core [3].

This reliance on the D-T reaction, however, creates a fundamental logistical challenge: the fuel cycle is not self-contained. While deuterium is abundant in seawater, tritium is a radioactive isotope with a short half-life of 12.32 years and is not found in nature in any significant quantity. The world’s current tritium supply is a by-product of heavy-water-moderated fission reactors, such as CANDU reactors. Multiple analyses of this supply chain have concluded that global stockpiles are extremely limited and face serious shortages [13, 14], *even without* the projected demand from a future fleet of fusion power plants.

Therefore, for fusion energy to be a sustainable and viable long-term power source, any commercial D-T reactor must breed its own tritium. This has led to the development of in-situ fuel production systems, known as **Tritium Breeder Blankets**.

3.2. Breeder Blanket: Concept, Function, and Evolution

A breeder blanket is a complex component surrounding the plasma-containing vacuum vessel. Its primary function is to intercept the 14.1 MeV fusion neutrons and use them to induce tritium-producing reactions in lithium. These are in reference to the two key breeding equations Eq. (4) and Eq. (5).

While ^7Li can breed, the ^6Li reaction is exothermic and possesses a much larger reaction cross-section for the thermalised neutrons dominant in a blanket, making it the primary breeding isotope.

The concept of a “breeder” blanket predates its application in fusion; it was first developed as a method to use fusion neutrons to produce fissile materials, such as ^{233}U and ^{239}Pu , for fission reactors [15, 16, 17]. Today, the design must also perform several other critical functions, including shielding the superconducting magnets from intense neutron radiation and extracting the fusion energy (via neutron thermalisation) for power conversion.

The key metric for a blanket’s performance is the **Tritium Breeding Ratio (TBR)**, defined as the ratio of tritium atoms produced in the blanket to the tritium atoms consumed in the plasma. Due to inevitable losses from neutron capture in structural materials, decay, and incomplete fuel burn-up, a TBR significantly greater than 1.0 is required to achieve tritium self-sufficiency [9, 10].

However, the D-T reaction produces only one neutron, and the ^6Li reaction consumes one neutron. This leaves no margin for losses. To achieve a $\text{TBR} > 1$, most blanket designs must include a **neutron multiplier** material. The most effective materials for this are beryllium (^9Be) and lead (^{208}Pb), which undergo (n, 2 n) reactions (spallation) when struck by high-energy fusion neutrons, effectively turning one incident neutron into two [7].

3.3. The Role of Neutron Multipliers

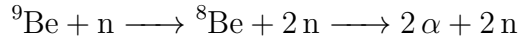
The D-T fusion reaction produces a single 14.1 MeV neutron, while the primary breeding reaction, $^6\text{Li}(n, \alpha)\text{t}$, consumes one neutron. This 1-to-1 neutron economy leaves no margin for

inevitable losses due to parasitic absorption in structural materials or neutrons escaping the blanket. To achieve the required Tritium Breeding Ratio (TBR) greater than unity, a blanket must therefore incorporate a **neutron multiplier** [12].

Natural lithium itself has a minor neutron-multiplying reaction via the high-energy (n, n't) reaction with ${}^7\text{Li}$, shown in Eq. (5). However this reaction is endothermic, consuming 2.5 MeV of energy, and its cross-section is insufficient on its own to guarantee tritium self-sufficiency [7]. This leads to a common consensus that dedicated neutron multipliers are required with the two primary candidates being Beryllium (Be) and Lead (Pb).

3.3.1. *Beryllium Neutron Multipliers*

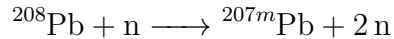
Beryllium, specifically the isotope ${}^9\text{Be}$, is an extremely effective neutron multiplier with a low energy threshold for its (n, 2 n) reaction:



It is widely used in conceptual solid breeder designs, such as the Helium-Cooled Pebble Bed (HCPB) and Water-Cooled Pebble Bed (WCPB), often in the form of beryllium pebbles or beryllide ceramics (Be_{12}Ti) [18, 19]. It is also a key component in the molten salt FLiBe ($2\text{LiF} \cdot \text{BeF}_2$) [20, 21]. However, beryllium has two major drawbacks: it is a scarce resource, and its dust is “ridiculously toxic” [22], posing significant safety, handling, and manufacturing challenges.

3.3.2. *Lead Neutron Multipliers*

Lead, typically natural lead or lead enriched in ${}^{208}\text{Pb}$, is the other main multiplier candidate. It undergoes a high-energy (n, 2 n) reaction:



Lead’s primary advantage is that it can be seamlessly integrated into a liquid breeder, forming the $\text{Li}_{17}\text{Pb}_{83}$ eutectic alloy. This allows the material to serve as breeder, multiplier, and coolant simultaneously [23, 12]. While it avoids the toxicity issues of beryllium, lead is a high-Z (high atomic number) material. This means it can also be a source of parasitic neutron absorption, particularly for the low-energy thermal neutrons that are most effective for breeding with ${}^6\text{Li}$. Furthermore, its activation by high-energy neutrons can lead to the production of long-lived radioisotopes, complicating waste disposal [24].

The choice of multiplier, therefore, represents a critical trade-off between neutronic efficiency, material toxicity, resource availability, and long-term waste management [7, 22].

3.4. Major Breeder Blanket Architectures

Blanket designs are broadly divided into two categories—liquid and solid—based on the phase of the lithium-bearing breeder material.

3.4.1. *Liquid Breeder Concepts*

Liquid breeders (LBs) are attractive because the breeding medium can simultaneously function as the coolant, simplifying the design and allowing for continuous tritium extraction outside the reactor [20].

3.4.1.1. *Lithium-Lead (Li–Pb):*

This is one of the most mature concepts, typically using the eutectic alloy $\text{Li}_{17}\text{Pb}_{83}$ [25, 23]. The lead acts as both a neutron multiplier and the primary component of the alloy, while the eutectic composition provides a low melting point, which is crucial for circulation [25]. This concept is the basis for the **Dual-Cooled Lithium-Lead (DCLL)** blanket, a primary candidate for the European DEMO reactor [26, 12]. A key challenge for all liquid metal blankets is magnetohydrodynamics (MHD), where the strong magnetic fields of the tokamak induce currents in the flowing metal, creating a drag force that inhibits circulation [12].

3.4.1.2. *Molten Salts:*

An alternative LB concept uses molten fluorine or chlorine salts. The most prominent example is **FLiBe** ($2\text{LiF} \cdot \text{BeF}_2$), which advantageously combines the breeder (LiF) and multiplier (BeF_2) into a single, low-conductivity fluid [20]. This is the reference design for the **Affordable Robust Compact (ARC)** reactor concept from MIT [21, 24, 27]. A wide variety of other salt compositions, including $\text{LiF} \cdot \text{PbF}_2$ and novel chlorine-based salts, are also under investigation to optimise breeding, temperature, and material compatibility [28, 29].

3.4.2. *Solid Breeder Concepts*

Solid breeders (SBs) use lithium-based ceramics, typically in the form of packed pebble beds, which eliminates MHD issues and offers high chemical stability [30, 31].

3.4.2.1. *Materials and Design:*

The leading candidate materials are lithium metatitanate (Li_2TiO_3) and lithium orthosilicate (Li_4SiO_4), often mixed with separate beryllium-based pebbles (e.g., Be_{12}Ti) as a multiplier [7, 32]. A review by [7] provides a comprehensive survey of these and other solid-phase options.

3.4.2.2. *Implementations:*

These materials form the basis of several major international designs. The **Chinese Fusion Engineering Test Reactor (CFETR)** is developing a **Water-Cooled Pebble Bed (WCPB)** blanket using $\text{Li}_2\text{TiO}_3/\text{Be}_{12}\text{Ti}$ pellets [19, 33]. European and Japanese DEMO concepts have focused on **Helium-Cooled Pebble Bed (HCPB)** designs, which are also being developed for ITER Test Blanket Modules [34, 18, 35].

The primary challenges for SB concepts are the complex engineering required for cooling (either with high-pressure helium or water) and the difficulty of extracting tritium, which must diffuse out of the solid ceramic and can be trapped in radiation-induced vacancies [36].

3.5. Neutronics Modelling and the Research Gap

Evaluating the TBR and other neutronic parameters of these complex designs is extremely difficult to do experimentally [37, 38]. While small-scale mockups provide crucial data on material properties and tritium release, they cannot replicate the full neutron spectrum and geometry of a power plant [37]. Consequently, the field relies heavily on computational modelling, particularly **Monte Carlo (MC) simulations**. The probabilistic nature of neutron transport—defined by scattering, absorption, and multiplication cross-sections—is ideally suited to the statistical sampling methods of MC codes [39, 5].

Modern neutronic toolchains are central to this research effort. The field increasingly utilises state-of-the-art, open-source Monte Carlo codes, which have been validated against other established codes and shown to accurately reproduce TBR and neutron flux calculations for fusion systems [40, 41, 42, 43]. A significant hurdle in this process is the management of complex reactor geometries. Accurately translating detailed Computer-Aided Design (CAD) models into a format usable by the neutronics code is a well-known challenge, often requiring significant manual effort [44] and geometric simplification [28].

A review of the literature reveals that decades of research have narrowed the focus to a small selection of “safe” materials (e.g., Li-Pb, Li_4SiO_4 , Li_2TiO_3) chosen not only for breeding but for their structural, chemical, and cost properties [11, 8]. This necessary pragmatism has left many other material combinations, particularly those in complex molten salt systems, comparatively unexplored [7].

However, this material-focused pragmatism is not the only factor limiting the current design space. A significant methodological gap exists. Past neutronic studies have typically been forced to choose between breadth and depth. On one hand, broad surveys like [7] provide an excellent “first principles” review of a wide range of breeder and multiplier materials. But to compare so many options, these studies must rely on simplified, often one-dimensional, models, which cannot capture the complex three-dimensional neutron transport effects within a realistic tokamak geometry.

On the other hand, studies focused on specific reactor concepts, such as the molten salt analysis for DEMO by [45] or the liquid breeder comparison for ARC by [28], provide vital, high-fidelity 3D analysis. However, their conclusions are inherently tied to the *fixed geometry* of that specific reactor design. These studies typically compare a handful of materials within a single “dedicated model,” fixing crucial parameters like blanket thickness and structural fractions. Consequently, they can answer “What is the best material for *this* design?” but not “What is the best *combination* of material and geometry?” For example, such studies often present TBR as a function of ^6Li enrichment, but do not explore how varying the blanket thickness or material composition ratios would fundamentally change those results.

This limited scope was, until recently, a computational necessity. Manually creating and simulating thousands of unique, complex 3D CAD models for a full parametric sweep was computationally intractable. Recent advances in scriptable, parametric modelling workflows have fundamentally lowered this barrier, making it feasible to automatically generate and distribute simulations for vast libraries of reactor designs. Work by [46] has demonstrated the power

of this new toolchain by modelling *several* distinct geometries. While this provides an excellent proof-of-concept, but as an unpublished paper this foundational work represents an initial step and leaves a significant opportunity for a more comprehensive and systematic exploration, particularly regarding material composition.

This project aims to fill this gap. By leveraging an integrated modelling workflow, we will conduct a broad, systematic investigation of TBR performance. This work builds upon previous studies by moving beyond the 1D-vs-3D or material-vs-geometry trade-off. We will explore the *coupled design space* by systematically and simultaneously varying key parameters that are often fixed: ^6Li enrichment, blanket thickness, and material composition ratios in a realistic 3D geometry. This systematic parameter sweep will provide a comprehensive dataset to identify optimal *combinations* of parameters, identifying design points that previous, more narrowly-focused studies may have overlooked.

4. METHODOLOGY

4.1. Modelling Neutron Transport: From Boltzmann to Monte Carlo

The neutronic performance of the breeder blanket is determined by the transport behaviour of the entire neutron population within the reactor geometry. This behaviour is formally governed by the **Boltzmann Transport Equation (BTE)**, a linear integro-differential equation representing particle balance in phase space [5].

The BTE stipulates that the time rate of change of the neutron density is equal to the net production (via sources and fission/multiplication) minus the net loss (via absorption and leakage) and the net scattering transfer into or out of the phase space volume. While the BTE provides a comprehensive deterministic description, it is computationally intractable to solve analytically for the complex, heterogeneous 3D geometries inherent to fusion reactors. Consequently, this project employs the **Monte Carlo (MC) method**.

The MC method is a stochastic technique that simulates the transport of neutrons by tracking the individual life histories of discrete particles [39]. Rather than solving for global flux directly, the simulation proceeds through a random walk process:

1. **Source Sampling:** A neutron is initialized at the plasma source with sampled parameters (e.g., 14.1 MeV energy and isotropic direction).
2. **Ray Tracing:** The distance to the next interaction is sampled from a probability distribution derived from the total macroscopic cross-section of the local material [6].
3. **Interaction Physics:** At the collision site, the specific reaction type (elastic scatter, absorption, $(n, {}_2n)$ multiplication, etc.) is selected probabilistically based on the nuclide-specific cross-section data.
4. **History Termination:** The particle's energy and direction are updated after scattering. The process repeats until the neutron is absorbed or leaks from the system geometry. Multiplication events generate secondary neutrons which are subsequently tracked.

4.1.1. *Statistical Uncertainty and Error Analysis*

Unlike deterministic solvers, Monte Carlo results represent statistical estimates rather than exact solutions. The physical quantities of interest, such as the Tritium Breeding Ratio (TBR), are tallied by averaging contributions across many independent particle histories. The uncertainty in these results is quantified using the standard deviation of the mean across simulation batches.

The relative error (R) in an MC calculation is governed by the Central Limit Theorem and scales inversely with the square root of the number of simulated particles [5, 40] (N):

$$R \propto \frac{1}{\sqrt{N}} \quad (6)$$

To ensure statistical robustness, the simulations in this work utilised high particle counts (typically $N \sim 10^5 - 10^7$) to drive convergence. This resulted in relative errors significantly below 1%, rendering the associated error bars smaller than the data markers in the presented graphical results.

4.2. Computational Workflow and Toolchain

As established in Section 3, modern neutronics research relies on a robust toolchain linking geometric modeling with particle transport simulation. This project employs a coupled workflow utilising two primary open-source tools: **Paramak** for geometry generation and **OpenMC** for neutron transport.

4.2.1. *Geometry Generation: Paramak*

To manage the complex toroidal geometry of the tokamak, this project utilises **Paramak**, a Python-based parametric CAD framework [44]. **Paramak** allows for the automated generation of tokamak fusion reactor models through a high-level API, enabling the rapid modification of geometric features—such as blanket thickness, triangularity, and radial build—without manual CAD reconstruction.

The generated geometry is exported as a DAGMC (Direct Accelerated Geometry Monte Carlo) unstructured mesh. This format is optimised for ray-tracing operations, allowing the Monte Carlo code to transport particles directly on the CAD surface representation without voxelization.

4.2.2. *Neutron Transport: OpenMC*

All neutronic simulations are performed using **OpenMC**, a state-of-the-art Monte Carlo transport code [40]. **OpenMC** is widely adopted within the fusion community due to its continuous-energy physics and efficient parallelisation. The code has been extensively validated for complex fusion applications, demonstrating high fidelity in calculating critical parameters such as the Tritium Breeding Ratio (TBR) and volumetric nuclear heating [41].

To ensure computational efficiency while maintaining physical accuracy, the simulations utilise a 90° sectoral revolution of the tokamak with reflective periodic boundary conditions applied to the radial surfaces (Figure 4).

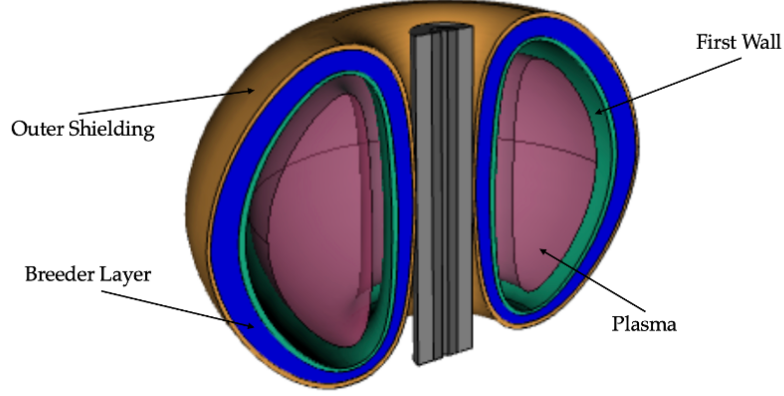


Figure 4: Representative geometry of the simplified Tokamak model generated via Paramak. The simulation domain comprises a 90° revolution with periodic boundary conditions to approximate the full torus while reducing computational cost.

4.3. Definition of the Base Reactor Model

Before undertaking parametric optimisation, a “Base Model” was established to serve as a reference point. The parameters for this baseline were synthesised from literature to ensure the reactor possesses dimensions and material compositions representative of a realistic DEMO-class machine. A summary of these base parameters is provided below.

4.3.1. Neutron Source Specification

The plasma source is modelled as an isotropic, 14.1 MeV neutron ring source located at the magnetic axis. To approximate the Doppler broadening resulting from the thermal motion of plasma ions, the energy spectrum follows a Muir distribution [47]. The ion temperature is set to 20 keV, consistent with burning plasma conditions described by Wandelt [48]. Note that the plasma is treated solely as a neutron ring source; a physical plasma volume with self and wall interactions are outside the scope of this neutronic study.

4.3.2. Breeder and Coolant Composition

The reference liquid breeder material is the eutectic Lithium-Lead alloy ($\text{Li}_{17}\text{Pb}_{83}$). This specific stoichiometry is selected for its low melting point, mitigating freezing concerns in the loop [25]. There is no hard consensus regarding Li_6 enrichment, literature varies from a low as 10% all the way to 90% [7, 45, 27, 28, 49]. To take an average values of the most common values found in breeding studies, the ^6Li enrichment is initialised at 60%.

Water (H_2O) is selected as the base coolant due to its prevalence in designs such as the ITER TBM and DEMO concepts [38]. Coolant Piping Setups differ from significantly between reactor design. To maintain computational tractability, the breeder and coolant are modelled as a homogenised mixture rather than discrete cooling channels. The coolant occupies 33% of the total blanket volume, as an estimation of piping volume of typical setups [18].

For the comparative analysis of solid ceramic breeders, a specific multiplier configuration was defined to ensure adequate neutron population, informed as a design common in a breadth of

older and more recent literature [11, 18, 31]. These simulations utilized a Titanium-Beryllide multiplier (Be_{12}Ti) mixed with the ceramic breeder at a Breeder-to-Multiplier volume ratio of 1:3. To accurately approximate the macroscopic density of a ceramic pebble-bed concept, a packing factor of 62% was applied to the material definition, accounting for the inter-particle void space inherent to packed beds.

4.3.3. *Reactor Geometry and First Wall*

The First Wall (FW) material is Zirconium Alloy 4 (Zr-4: 98.2% Zr, 1.5% Sn, 0.2% Fe, 0.1% Cr). Zirconium is selected over standard ferritic-martensitic steels for the base model due to its significantly lower neutron absorption cross-section [43]. The FW thickness is set to 25 mm, an average value derived from structural integrity studies [45, 33].

The breeder blanket depth is set to an average of 500 mm, consistent with modular designs found in literature [33, 19]. The reactor triangularity is fixed at 0.55 in the base model.

4.4. **Simulation Strategy and Parametric Sweeps**

To explore the design space surrounding the Base Model, this project implements a custom, automated simulation pipeline. Leveraging the Python interfaces of both **Paramak** and **OpenMC**, an iterative script varies specific design parameters while holding the remaining Base Model attributes constant.

The workflow for each parametric sweep follows this sequence:

1. **Parameter Injection:** A target variable (e.g., Blanket Thickness, ^6Li Enrichment, or Coolant Fraction) is modified in the input dictionary.
2. **Geometry Reconstruction:** **Paramak** regenerates the 3D CAD model and converts it to the DAGMC h5m format.
3. **Material Assignment:** **OpenMC** applies the material definitions, updating isotopic fractions (e.g., adjusting the lead-lithium ratio) as required by the sweep.
4. **Transport Simulation:** The stochastic transport simulation is executed, tallying neutron interactions in the blanket volume to determine the TBR.

This methodology allows for the isolation of individual variable effects, providing clear trends on how geometric and material changes influence the overall neutronic performance of the reactor.

5. **RESULTS**

5.1. **Material Composition and Performance**

The neutronic performance of candidate liquid and solid breeder materials is summarized in Table 1 and Table 2, respectively.

Among the liquid breeders, the eutectic Lithium-Lead ($\text{Li}_{17}\text{Pb}_{83}$) alloy demonstrates the highest TBR of 1.61. The distinctive factors to this superior performance are attributed to

Table 1: A summary of liquid breeder materials, their properties, and simulated TBR (error on each ≤ 0.005) and simulated Average Heating per Neutron (error on each ≤ 0.05).

Material	Ratio (molar %)	Li (molar %)	ρ (g/cm ³)	TBR	Heating (MeV)
Li ₁₇ Pb ₈₃	-	17.00	11.00	1.61	11.54
LiF–BeF ₂	2:1	28.57	2.04	1.15	12.92
LiF–BeF ₂ *	1:1	20.00	2.06	1.16	12.97
LiF–PbF ₂	2:3	15.38	3.55	1.23	11.63
LiF	-	50.00	2.64	1.21	13.49
Li	-	100	0.51	1.17	13.60
Li ₃ N	-	75.00	1.30	1.13	13.33
LiF–BeF ₂ –NaF	1:1:1	14.29	2.15	1.09	12.42
LiF–NaF–ZrF ₄	55:22:23	20.45	2.72	1.07	11.59
LiF–LiI ₂	83.5:16.5	50.00	3.68	1.02	10.96
LiF–LiBr–NaF	14:79:7	45.69	3.20	0.98	15.57
LiF–LiBr–NaBr	20:73:7	46.50	3.16	0.97	15.55
LiCl–PbCl ₂	1:2	12.50	4.50	0.96	9.87
LiCl–NaCl	72:28	36.00	1.52	0.91	11.04
LiF–NaF–KF	46.5:11.5:42	20.17	2.02	0.90	11.33
LiCl–BeCl ₂	1:1	20.00	1.52	0.88	10.90
LiCl–KCl	7:3	35.00	1.51	0.85	10.72
LiCl–KCl*	52:48	26.00	1.51	0.80	10.37

higher density and high lead multiplier fraction. The density arises as it is an eutectic alloy that can utilise higher packing efficiency [46, 27]. While high TBR typically correlates with increased nuclear heating due to the exothermic ${}^6\text{Li}$ reaction, $\text{Li}_{17}\text{Pb}_{83}$ is a notable outlier. It exhibits lower-than-expected heating relative to its TBR because the neutron multiplication reactions in lead are endothermic.

Molten salts generally result in lower TBRs compared to liquid metals but are investigated due to their reduced Magnetohydrodynamic (MHD) effects. Of the halide salts, Fluorine-based compounds (e.g., FLiBe) perform best due to their low neutron absorption cross-sections. While Chlorine and Bromine salts were analyzed, Bromine salts exhibit high parasitic absorption, leading to significantly higher heating values (approx. 15.5 MeV) despite lower tritium production. However, it is noted that with realistic First Wall thicknesses, Bromine outperforms Chlorine due to its moderating properties.

Pure lithium, LiF and LiN_3 (and to a lesser extent LiF-LiI_2) have significant Li-7 TBR contributions due to the lack of a dedicated breeder element within the compound.

Table 2: A summary of solid breeder materials, their properties, and simulated TBR (error on each ≤ 0.005) and simulated Average Heating per Neutron (error on each ≤ 0.05).

Material	Li (molar %)	ρ (g/cm ³)	TBR	Heating (MeV)
Li_4SiO_4	44.44	2.40	1.47	14.88
Li_2TiO_3	33.33	3.43	1.46	14.63
Li_8PbO_6	53.33	4.28	1.51	15.01
Li_8CeO_6	53.33	3.25	1.51	15.29
Li_8ZrO_6	53.33	2.98	1.50	14.99
Li_8SnO_6	53.33	3.41	1.49	15.18
$\text{Li}_6\text{Zr}_2\text{O}_7$	40.00	3.56	1.49	14.79
Li_8GeO_6	53.33	2.64	1.49	14.97
Li_5FeO_4	50.00	2.64	1.48	14.91
Li_5AlO_4	50.00	2.25	1.48	15.00
Li_8SiO_6	53.33	2.20	1.48	15.03
Li_6ZnO_4	54.55	2.86	1.48	14.95
Li_4TiO_4	44.44	2.57	1.48	14.89
Li_6MnO_4	54.55	2.50	1.48	14.91
Li_4GeO_4	44.44	3.16	1.48	14.79
Li_2O	66.66	2.10	1.47	14.92
Li_2SiO_3	33.3	2.53	1.45	14.75
Li_2MnO_2	40.00	3.90	1.43	14.42
Li_8CoO_6	53.33	2.47	1.43	14.69
Li_6CoO_4	54.55	2.77	1.41	14.58

Table 2 indicates that solid ceramics display remarkably consistent performance compared to liquids, displayed by the low standard deviation on the average TBR: $\text{TBR} = 1.48 \pm 0.03$.

This uniformity arises both from the relatively similar atomic compositions of the molecules (mostly lithium and oxygen) and the standardised multiplier configuration (Be_{12}Ti at a 1:3 ratio) defined in Section 4.3.2.

Among the specific compounds, Cerium-based ceramics achieved the highest TBR due to the low parasitic capture cross-sections of Cerium. Conversely, Cobalt-based compounds act as outliers, yielding lower TBRs despite the high multiplier fraction; this is attributed to the higher absorption and scattering cross-sections of Cobalt, which degrade the neutron economy.

5.2. Isotopic and Compositional Optimisation

Figure 5 illustrates the sensitivity of TBR to ^6Li enrichment. The dip in TBR at high percentages is attributed to the complete depletion of ^7Li , eliminating the contribution of $^7\text{Li}(n, n'\alpha)t$ reactions which are valuable for preserving the neutron population in the high-energy spectrum.

It is noted that the enrichment level of 60% used in the base model was selected *a priori* based on literature standards. However, the simulation data indicates that this value lies well within the saturation plateau, suggesting that the neutronic maximum is reached at significantly lower enrichment percentages.

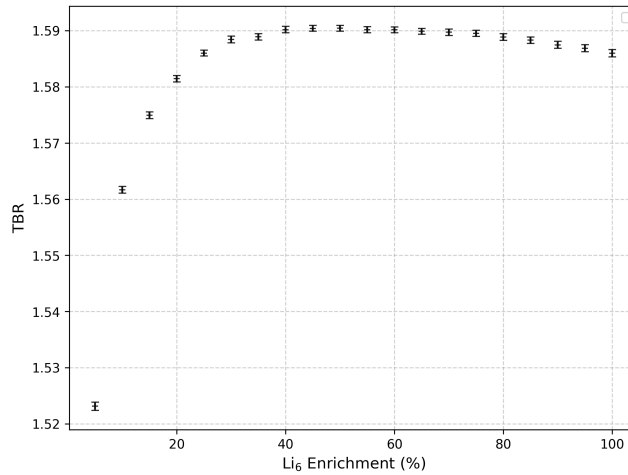


Figure 5: Tritium Breeding Ratio (TBR) as a function of ^6Li enrichment percentage. The curve is characterised by a rapid initial increase which saturates early; the “knee” of the curve appears at approximately 20%, with the TBR effectively capping at 40%. Beyond this saturation point, the returns are diminishing. Notably, a slight suppression in breeding performance is observed at near-total enrichment levels ($> 90\%$)

The simplified study of the lead multiplier fraction in LiPb breeders is shown in Figure 6, where an approximation was taken where the density is calculated as a simple weighted average of Li and Pb. Though there was a peak at 50% lead composition, this TGR was still far lower than the TBR from $\text{Li}_{17}\text{Pb}_{83}$. These results show that density is a stronger factor affecting liquid TBR improvements than simple multiplier - breeder ratios.

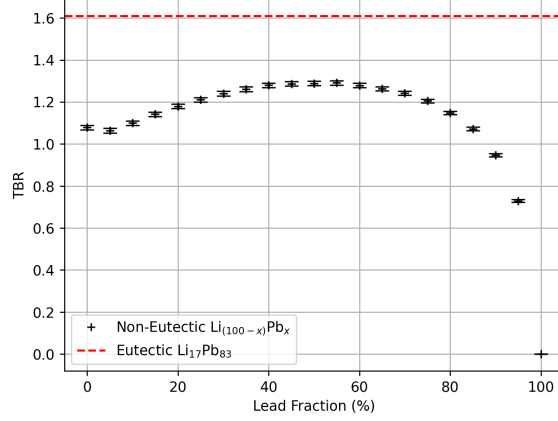


Figure 6: TBR dependence on Lead fraction in a simplified non-eutectic / constant packing factor $\text{Li}_{(100-x)}\text{Pb}_x$ breeder, where the density is calculated as a simple weighted average of Li and Pb. Breeding performance peaks at approximately 50% lead concentration, with a rapid decline observed beyond 75%. The eutectic point is marked for reference, far exceeding any non-eutectic TBR. Error bars are included but are too small to be visible at this scale.

5.3. Impact of Coolant Selection

The influence of coolant fraction and type on breeding efficiency is detailed in Figure 7. For Helium and Nitrogen, TBR decreases monotonically as the coolant fraction increases. This behaviour indicates that these gases effectively act as voids within the breeder blanket, displacing active breeding material.

Conversely, Water (H_2O) exhibits a non-monotonic trend where TBR initially increases before declining. Notably the TBR was the very similar at 40% water coolant as when no coolant was simulated. This is attributed to the moderating properties of water, which thermalises the neutron spectrum. As shown in Figure 7b, the water coolant shifts the incident neutron population toward lower energies. Since the tritium production cross-section of ${}^6\text{Li}$ is significantly higher at thermal energies, this moderation compensates for the displacement of breeder volume at lower coolant fractions. This moderation will also reduce the neutron leakage, and so improve heating along with TBR. This then introduces the importance of considering neutron mean free path / flux attenuation and leakage within the blanket and so leads us towards the geometric studies.

5.4. Geometric Parameter Sweeps

Results from the geometric optimisation studies indicate that reactor triangularity has a negligible impact on global neutronics (Figure 8), which is expected given the uniform ring source approximation.

However, the First Wall (FW) thickness acts as a significant driver of performance. Figure 9 shows a linear decrease in TBR as FW thickness increases, caused by parasitic absorption in the structural material. A Material comparison reveals that while EUROFER [50] is the standard structural choice, Zirconium-4 alloy provides superior neutronic performance, as they have the

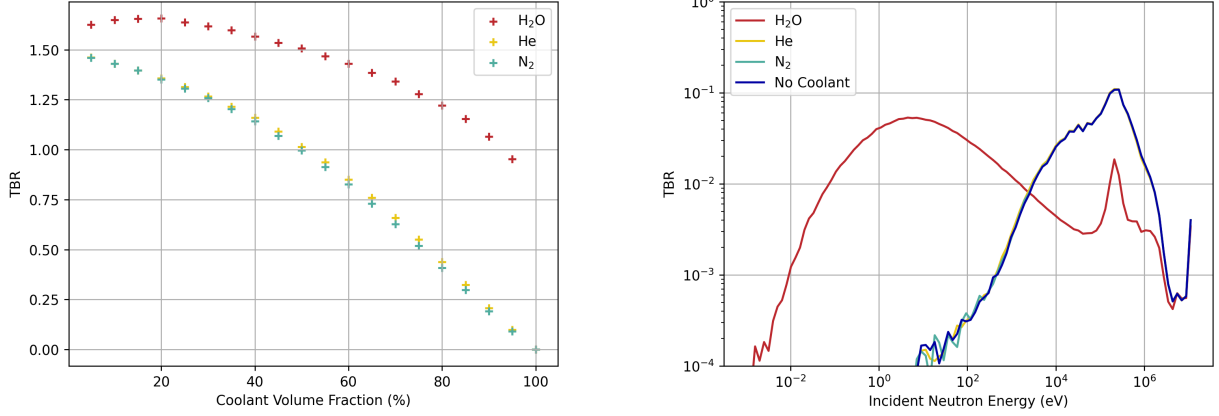


Figure 7: Impact of coolant choice on breeding performance. (a) TBR vs. Fraction of Breeder Blanket Layer replaced with Coolant (by volume). Gaseous coolants (He, N) act as voids and cause steady reductions in TBR as fractions increase. H₂O, however, has a relatively flat TBR curve up to a fraction of 0.5, even increasing TBR at low fractions. The TBR had shallower decreases at high water fractions. Error bars are included but are too small to be visible at this scale. (b) Incident Neutron Energy spectrum for Coolant Volume Fraction = 0.33. The H₂O spectrum shows significant moderation compared to the gaseous coolants, increasing the interaction cross-section.

lowest parasitic neutron absorption.

Finally, Figure 10 determines the optimal radial depth of the breeder blanket. TBR increases rapidly with thickness before saturating in the 400–600 mm range. This saturation suggests that the neutron flux is fully attenuated beyond this depth, a conclusion supported by the reaction density heatmap (Figure 10b). Increasing the breeder depth beyond 600 mm yields negligible gains in tritium production.

6. DISCUSSION

The results presented in Section 5 quantify the neutronic performance of various breeder blanket concepts. While the simulation data identifies specific maxima for Tritium Breeding Ratio (TBR), the selection of a viable blanket design for a fusion power plant requires a holistic evaluation that balances neutronic efficiency against thermal-hydraulic, structural, and economic constraints [7, 34].

6.1. TBR Optimisation Strategies

6.1.1. Isotopic and Compositional Efficiency

The study confirms that the eutectic Lithium-Lead alloy (Li₁₇Pb₈₃) offers superior breeding performance compared to molten salts, primarily driven by the neutron multiplication capabilities of lead [12]. However, the simulation highlights a critical physical distinction: the superior performance of the eutectic alloy over non-eutectic mixtures (Figure 6) cannot be explained by stoichiometry alone. The eutectic formation allows for a packing efficiency that results in

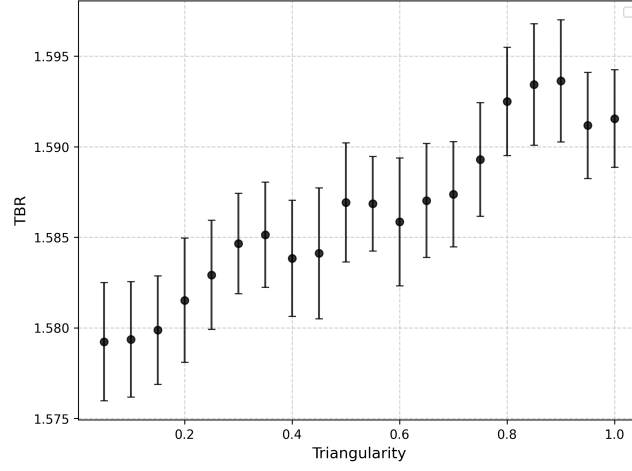


Figure 8: Sensitivity of TBR to reactor triangularity (δ). The gradient of 0.015 ± 0.001 indicates that triangularity has a negligible impact on global neutronics, consistent with the use of a uniform ring source model.

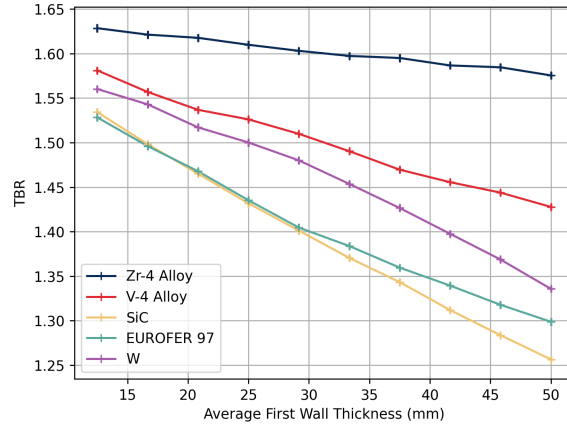


Figure 9: TBR degradation as a function of First Wall (FW) thickness for various candidate materials, including EUROFER97 [50] and Zr-4 as defined in Section 4.3.2. Zirconium-based alloys demonstrate the lowest parasitic absorption, maintaining the highest TBR profile. Error bars are included but are too small to be visible at this scale.

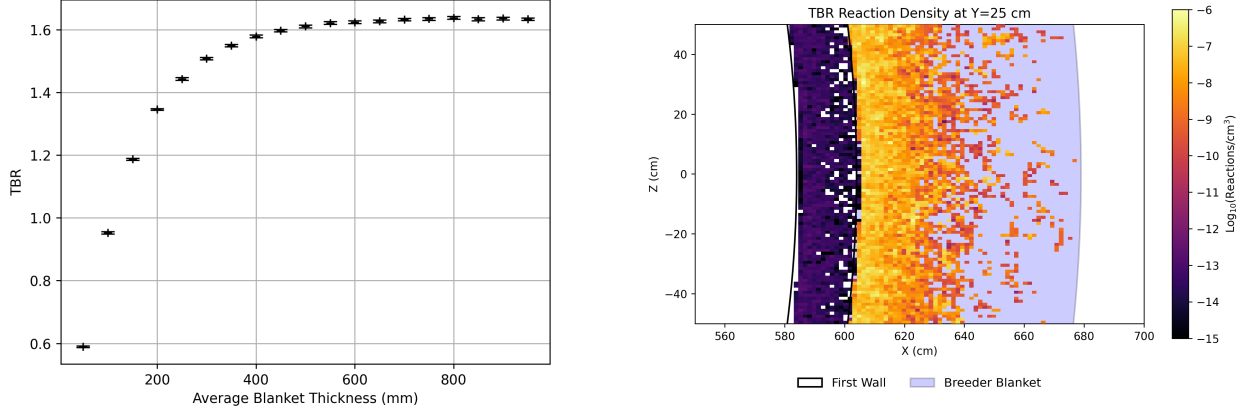


Figure 10: Analysis of Breeder Blanket radial thickness. (a) TBR saturates at approximately 500 mm, suggesting full neutron attenuation. (b) Cross-sectional heatmap of reaction density. The reaction rate drops to negligible levels beyond the 600 mm mark, confirming the saturation thickness observed in (a). Error bars are included but are too small to be visible at this scale.

a density approximately 15% higher than the weighted average of its constituents [25]. This increased macroscopic cross-section is the dominant factor in maximising the neutron economy.

A significant finding for reactor optimisation is the behaviour of ^6Li enrichment (Figure 5). While high enrichment is often assumed to be beneficial, our results indicate a saturation point at approximately 40% enrichment. Beyond this point, the blanket becomes “neutron-limited” rather than “target-limited.” Furthermore, enrichment levels approaching 100% actually degrade performance. This is attributed to the total depletion of ^7Li , which removes the contribution of the $^7\text{Li}(n, n'\alpha)t$ reaction [9]. Although this reaction is less frequent, it preserves the neutron population by emitting a secondary neutron, whereas the ^6Li reaction is purely absorptive. Economically, this suggests that costly high-enrichment processes are unnecessary; a moderate enrichment of 20–40% represents the optimal trade-off between TBR and fuel cycle costs [46, 27].

6.1.2. Spatial and Geometric Optimisation

The saturation of TBR at a radial depth of roughly 500 mm (Figure 10) can be validated physically by considering the neutron mean free path (λ). Dominated by lead, the macroscopic total cross-section (Σ_t) of the breeder mixture yields a mean free path of $\lambda \approx 5.5$ cm for 14 MeV neutrons. A 500 mm blanket therefore represents approximately 9 mean free paths, sufficient to attenuate over 99.9% of the incident flux [5]. Increasing the blanket thickness beyond this point provides no neutronic benefit and would act only as dead weight, complicating remote maintenance and increasing capital costs [8].

6.2. Consequences to Other Critical Criteria

While this study maximised TBR, the “optimal” neutronic configurations often conflict with other critical reactor criteria.

6.2.1. Structural Integrity vs. Neutronics

The use of Zirconium-4 (Zr-4) as the First Wall (FW) material yielded the highest TBR due to its low parasitic absorption cross-section (Figure 9) [43]. However, Zr-4 is generally unsuitable for the extreme neutron fluence and thermal stress of a DEMO-class reactor FW. Standard Reduced Activation Ferritic-Martensitic (RAFM) steels, such as EUROFER 97, are the engineering standard despite their higher neutron penalty [50]. Our results suggest that transitioning from an idealised Zr-4 wall to a realistic steel wall results in a TBR penalty of approximately 0.1 to 0.15. This parasitic loss must be compensated for by increasing the breeder volume or enrichment, highlighting the tight margins in fusion neutronics [34].

6.2.2. Safety and Coolant Selection

The choice of coolant represents a stark trade-off between performance and safety. Water (H_2O) was shown to enhance TBR (Figure 7) significantly better than gaseous coolants due to its moderating power, which softens the neutron spectrum into the thermal region where the ^6Li cross-section is largest [49]. However, the use of water in a lithium-based blanket introduces the risk of violent exothermic chemical reactions in the event of a leak [35, 38]. While Helium avoids this chemical reactivity, our results show it acts effectively as a void, increasing neutron leakage and degrading TBR [18]. A realistic design must therefore weigh the neutronic advantage of water moderation against the engineering complexity of double-walled piping and pressure suppression systems required to mitigate chemical hazards [19].

6.2.3. Energy Extraction (Heating)

A counter-intuitive trade-off exists regarding nuclear heating. While $\text{Li}_{17}\text{Pb}_{83}$ provided the highest TBR, it exhibited lower average heating per neutron compared to ceramic breeders (Table 1 vs Table 2). This is intrinsic to the physics of the multiplier: the $(n, 2n)$ reaction in lead is endothermic, consuming incident energy to liberate neutrons [51]. Conversely, the parasitic absorption in materials like Bromine salts yields high heating but low breeding [29]. For a commercial power plant, where electricity generation is as vital as fuel self-sufficiency, the lower energy multiplication factor of lead-lithium systems is a distinct disadvantage that must be factored into the overall plant efficiency calculations [10].

6.3. Model Limitations

The primary limitation of this study lies in the homogenisation of the blanket materials. By treating the breeder and coolant as a uniform mixture, the simulation neglects “streaming effects,” where neutrons escape through the void spaces of coolant channels without interaction [42]. This likely results in a slight overestimation of the TBR in our results compared to a heterogeneous model.

Furthermore, the negligible impact of triangularity on TBR (Figure 8) is likely an artifact of the simplified neutron source definition. The model utilised a uniform ring source which does not geometrically conform to the plasma shape variations associated with high triangularity [44]. In a physical tokamak, increasing triangularity compresses the plasma volume and alters

the view factor of the neutrons incident on the first wall, an effect not captured by the current source definition [52].

6.4. Next Steps

Future work will focus on increasing the fidelity of the simulation model to address the limitations identified above.

- **Heterogeneous Geometry:** Utilising the advanced features of Paramak to model discrete cooling channels and pebble beds to quantify the effects of neutron streaming [44].
- **Plasma Source Definition:** Replacing the ring source with a parametric plasma source that accurately maps the spatial distribution of fusion events, allowing for a true investigation into the effects of plasma geometry [53].
- **Structural Integration:** Moving beyond the idealised Zr-4 FW to simulate a full CAD-based model of a DEMO-like blanket module, including internal stiffening plates and manifolds, to provide a more conservative and realistic estimate of the global TBR [8, 33].

7. CONCLUSION

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